Dayoon Ko

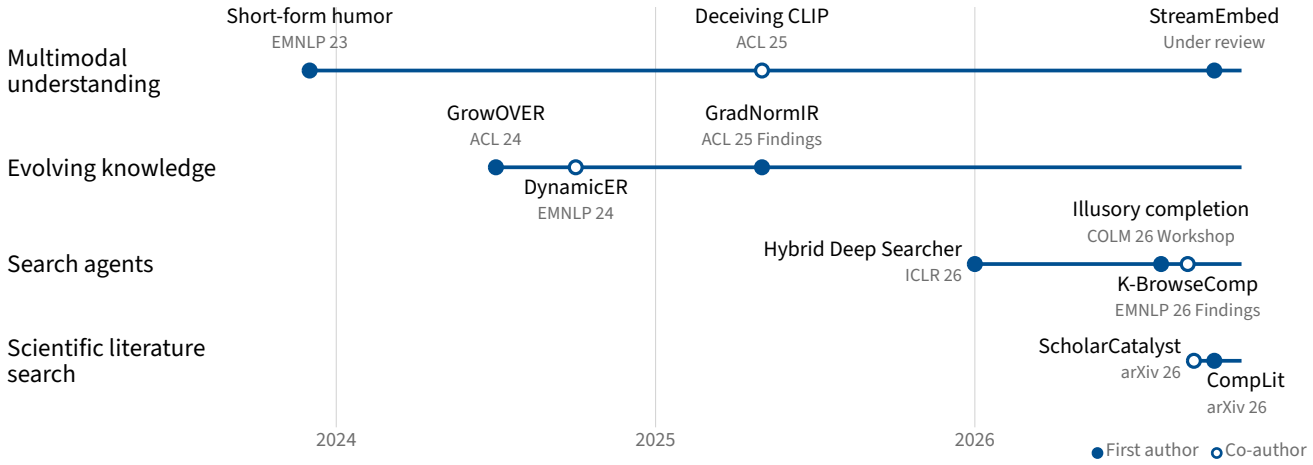
Berkeley, CA, USA dayoon.ko@vision.snu.ac.kr (650) 444-5010 dayoon-ko.github.io
dayoon-ko Google Scholar dayoon-ko

Summary

PhD candidate in Computer Science and Engineering at Seoul National University, advised by Gunhee Kim.

Research interests: how language models use external knowledge, including search agents, scientific literature search, retrieval-augmented generation (RAG) for evolving knowledge, and multimodal understanding.

Research Map



Education

PhD **Seoul National University**, Computer Science and Engineering — Seoul, Korea, Republic of
M.S./Ph.D. integrated student at the Vision & Learning Lab. — Sept 2022 – present
- Advised by Gunhee Kim.
- Research on search agents, scientific literature search, and RAG for evolving knowledge.

BS **Yonsei University**, Computer Science and Engineering — Seoul, Korea, Republic of
Bachelor of Science in Computer Science and Engineering. — Mar 2018 – Aug 2022
- GPA: 4.12/4.30, ranked first in the department (1/28).

Experience

UC Berkeley, Visiting Researcher — Berkeley, CA, USA
Hosted by Sewon Min. — Oct 2026 – present
- Developed StreamEmbed, an incremental visual embedding system for real-time screenshot indexing.

LG AI Research, Superintelligence Lab, Research Intern — Seoul, Korea, Republic of
Worked on agentic reasoning and retrieval systems for large language models. — Mar 2025 – Sept 2026
- Developed Hybrid Deep Searcher, a search agent that integrates parallel and sequential search reasoning (ICLR 2026).
- Analyzed how search agents conclude tasks with unverified constraints (COLM 2026 Workshop).
- Built CompLit, a scientific literature search benchmark for implicit, cumulative, and unmet needs.

Publications

StreamEmbed: Real-time Indexing from Pixels for Screenshot Retrieval Oct 2026
Dayoon Ko, Yichuan Wang, Soyeong Jeong, Joseph E. Gonzalez, Gunhee Kim, Sewon Min
Under review

CompLit: Scientific Literature Search Benchmarks Must Cover Implicit, Cumulative, and Unmet Needs Oct 2026
Dayoon Ko, Jihyuk Kim, Soyeong Jeong, Young-Jun Lee, Dahyun Lee, Juyeon Kim, Gunhee Kim, Moontae Lee, Kyungjae Lee
arxiv.org/abs/2610.28658 (arXiv 2026)

ScholarCatalyst: A Benchmark for Retrieving Papers That Inspire New Research Oct 2026
Sohyeon Kim, Yoonho Lee, Bo Liu, Dayoon Ko, Rulin Shao, Seungone Kim, Graham Neubig, Pang Wei Koh, Aakanksha Chowdhery, Akari Asai, Omar Khattab, Yejin Choi, Gunhee Kim, Chelsea Finn
arxiv.org/abs/2610.02202 (arXiv 2026)

K-BrowseComp: A Web Browsing Agent Benchmark Grounded in Korean Contexts Sept 2026
Nahyun Lee, Dongkeun Yoon, Guijin Son, Geewook Kim, Dayoon Ko, Jeonghun Park, Haneul Yoo, Jaewon Cho, Junghun Park, Changyoon Lee, Kyochul Jang, Jaeyeon Kim, Eunsu Kim, Woojin Cho, Seungone Kim
arxiv.org/abs/2606.02404 (EMNLP 2026 Findings)

When Is Enough Not Enough? Illusory Completion in Search Agents Aug 2026
Dayoon Ko, Jihyuk Kim, Sohyeon Kim, Haeju Park, Dahyun Lee, Gunhee Kim, Moontae Lee, Kyungjae Lee
arxiv.org/abs/2602.07549 (COLM 2026 Workshop on Agent Behavior)

Hybrid Deep Searcher: Scalable Parallel and Sequential Search Reasoning Jan 2026
Dayoon Ko, Jihyuk Kim, Haeju Park, Sohyeon Kim, Dahyun Lee, Yongrae Jo, Gunhee Kim, Moontae Lee, Kyungjae Lee
openreview.net/forum?id=rXpTZyucal (ICLR 2026)

When Should Dense Retrievers Be Updated in Evolving Corpora? Detecting Out-of-Distribution Corpora Using GradNormIR May 2025
Dayoon Ko, Jinyoung Kim, Sohyeon Kim, Jinhyuk Kim, Jaehoon Lee, Seonghak Song, Minyoung Lee, Gunhee Kim
aclanthology.org/2025.findings-acl.1334 (ACL 2025 Findings)

Can LLMs Deceive CLIP? Benchmarking Adversarial Compositionality of Pre-trained Multimodal Representation via Text Updates May 2025
Jaewoo Ahn, Heeseung Yun, Dayoon Ko, Gunhee Kim
aclanthology.org/2025.acl-long.1280 (ACL 2025)

DynamicER: Resolving Emerging Mentions to Dynamic Entities for RAG Oct 2024
Jinyoung Kim, Dayoon Ko, Gunhee Kim
aclanthology.org/2024.emnlp-main.762 (EMNLP 2024)

GrowOVER: How Can LLMs Adapt to Growing Real-World Knowledge? July 2024
Dayoon Ko, Jinyoung Kim, Hahyeon Choi, Gunhee Kim
aclanthology.org/2024.acl-long.181 (ACL 2024)

Can Language Models Laugh at YouTube Short-form Videos? Dec 2023
Dayoon Ko, Sangho Lee, Gunhee Kim
aclanthology.org/2023.emnlp-main.176 (EMNLP 2023)

Honors

Yulchon AI Star Fellowship 2026
Yulchon Foundation and Seoul National University AI Institute.

Graduation Exhibition Excellence Award Feb 2022
Yonsei University, Computer Science. Project on mitigating data imbalance in image classification.

Academic Excellence Award Fall 2021
Yonsei University, for semester grades.

Udeok Scholarship Feb 2019 – Aug 2022
Academic excellence scholarship from the Udeok Foundation, with full tuition during undergraduate years.

Academic Activities

Teaching Assistant

- Computer Vision (Fall 2024)
- Theory and Lab of IoT, AI, and Big Data (Fall 2023)
- Probabilistic Graphical Models (Spring 2023)

Conference Reviewer

- ACL Rolling Review (October 2024)

Skills

Programming: Python, PyTorch, Java, C++; Git; large-scale experiment management

Languages: Korean (native), English (intermediate)